12.7.1 Mode of Transmission

Reproductive tract infections are caused by pathogenic bacteria, parasite, virus. It is mainly caused by pathogens entering into the body through the mucous membranes during unprotected vaginal, oral, anal intercourse with an infected partner. In developing countries bacterial infections like Gonorrhoea, Chlamydia, Syphilis, Bacterial Vaginosis, Lymphogranuloma Venereum, Trichomoniasis, Chancroid, and viral infections caused by Human Papilloma Virus, Hepatitis B Virus, Herpes Simplex Virus, Human Immunodeficiency Virus are very common.

12.7.2 Normal Flora of Reproductive Tract

Mycobacterium smegmatis, a harmless commensal found in the smegma of the genitalia of both men and women. In nomal men aerobic and anaerobic bacteria, lactobacilli, alpha haemolytic *Streptococci, Chalmydia trachomatis* and *Ureaplasma urealyticum* may also be present.

The adult female genital tract has a very complex microflora. The character of the population changes with the variation of the menstrual cycle. Mostly the predominant bacteria are acid tolerant *Lactobacilli*. Glycogen is accumulated in the vaginal wall due to ovarian hormonal activity. The breakdown of glycogen by the lactic acid bacteria (Doderlien's bacillus) leads to the formation of acidic pH (4.4-4.5). This acidic nature prevents the vagina from bacterial vaginosis and yeast infections. However before puberty and after menopause there is no glycogen formation. The normal flora during this period contain normal skin microorganisms. The vaginal pH is mild alkaline. The normal vaginal flora often includes *Listeria,* anaerobic *Streptococci, Mycoplasma, Gardnerella vaginalis, Neisseria, Spirochetes, Candida, Staphylococcus epidermidis.*

12.7.3 Pathogenesis

After the entry of pathogenic organisms, with sufficient incubation time, symptoms are clearly manifested in the affected individual. The most common symptoms include unusual vaginal discharge, penile discharge, pelvic pain, itching, abnormal or heavy vaginal bleeding, rashes, warts, lesions, burning or pain during urination. However most of the infections are asymptomatic, which act as a effective control of reproductive tract infections. Diseases of reproductive system are listed in Table 12.9.

Infobits

Tamilnadu has AIDS testing centres at all district head quarters with more than 55 Anti Retroviral Therapy(ART) centres and 750 (ICTC)-Integrated (voluntary) and confidential counselling and testing centres under the national AIDS control programme at district level government hospitals and medical colleges across the state.

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